

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) |       | Acceleration (gradient) or equivalent = $\frac{(v-u)}{t}$ (1)<br><br>Displacement (area) = $ut + \frac{1}{2}t(v-u)$ or $\frac{(u+v)}{2}t$ (1)<br><br>Substitution and clear algebra seen e.g. $x = ut + \frac{1}{2}t(at)$ seen (1)<br>No reference to graph award a maximum of 2 marks                                                                                                                                                                                                                                                                                                                                                                            | 3               |     |     | 3     | 3     |      |
|          | (b) | (i)   | $11 \sin 50^\circ [= 8.4 \text{ [3] m s}^{-1}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 | 1   |     | 1     | 1     |      |
|          |     | (ii)  | Attempt at substituting into $H = ut + \frac{1}{2}at^2$ even if incorrect signs (1)<br><br>$H = 8.4 \times 1.95 - \frac{1}{2} \times 9.81 \times (1.95)^2$ seen (1) Award 2 marks if only this seen<br><br>$H = -2.27 \text{ [m]}$ (1) accept 2.21 [m]<br>Accept use of $v = 8 \text{ m s}^{-1}$ leading to $H = 3.05 \text{ [m]}$ .<br><br><b>Alternative solution:</b><br><br>Time to reach max height = $\frac{(0-8.4)}{-9.81}$ (or 0.86 s) (1)<br><br>Time from $H$ to floor = $1.95 - (2 \times 0.86)$ (or 0.232 s) (1)<br>$H = (8.4 \times 0.23) + \frac{1}{2} \times 9.81 \times (0.23)^2$ (or 2.21 m) (1)<br>Consider carefully consequence of 'rounding' | 1<br>1          | 1   |     | 3     | 3     |      |
|          |     | (iii) | Horizontal component of velocity = $11\cos 50^\circ [= 7.1 \text{ m s}^{-1}]$ or by Pythagoras (1)<br><br>Substitution into $v_H = \frac{R}{t}$<br><br>$R = 7.1 \text{ (ecf)} \times 1.95 = 13.8 \text{ [m]}$ (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 | 2   |     |       | 2     |      |

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks available |          |          |           |          |          |
|----------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          | (c) |  | The time for which the shot is in the air will increase (Steve correct here - no explanation needed) (1)<br>However, increasing the angle will decrease the horizontal component of velocity (1)<br>Reasonable conclusion based on one or both factors<br>e.g. both of these factors will tend to cancel each other out, so not a good idea to follow Steve's advice or it is a bad idea because the angle needs to be decreased (1) |                 |          | 3        | 3         |          |          |
|          |     |  | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                                                                              | <b>5</b>        | <b>4</b> | <b>3</b> | <b>12</b> | <b>9</b> | <b>0</b> |